

Wenchen Han

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EDUCATION

University College London

Sept 2022 – (Exp.) Sept 2026

- Ph.D. candidate, Dept. of Computer Science. Supervisors: [Ran Ben Basat](#) [🔗](#), [Brad Karp](#) [🔗](#)

Peking University

Sept 2018 – June 2022

- B.S. in Intelligence Science and Technology, Turing Class.
- GPA: 3.791/4.0, Rank: 4/82, Major GPA: 3.814/4.0
- Thesis: Hierarchical Aggregation for Efficient and Scalable Federated Learning in WAN.

PROFILE

Systems Ph.D. candidate working on algorithm-system co-designs for LLM infrastructure / distributed ML systems. My recent work focuses on large-scale LLM training and inference systems, including compressed all-reduce for distributed fine-tuning, KV-cache transfer for disaggregated serving, and MoE expert offloading under constrained GPU memory.

- Algorithm-system co-designs for LLM training and inference infrastructure; vLLM/PyTorch/NCCL/CUDA; LLM deployment, Machine learning; Performance optimization.

SELECTED RESEARCH AND WORK EXPERIENCES

Research Intern, Lynx: Accelerating KV Transfer for LLM Inference

Edinburgh, UK

SIR team, HW R&D UK. Manuscript C.

June 2025 - Jan 2026

- Designed Lynx, a split-stream KV-cache transfer system for disaggregated large-scale LLM inference for serving.
- Developed progressive/speculative KV transfer mechanism to overlap transfer with decoding under partial KV state.
- Implemented NPU kernels and integrated into vLLM serving pipeline, achieving $1.43\times$ speed-up and BF16 accuracy.
- Collaborated with another Huawei's engineering team to deploy Lynx in production.

Research Assistant, Gradient Quantization for DDP LLM Training

University College London

DynamiQ. Publication A and B at HotNets '24 and SIGCOMM '26.

Mar 2023 – June 2025

- Developed an **end-to-end distributed LLM fine-tuning pipeline** via a custom gradient-compression architecture.
- Maximized multi-node scalability by co-designing custom quantization with all-reduce network topologies.
- Built a high-performance, fully pipelined **C++/CUDA** and **NCCL** + my own **RDMA** backend, smoothly integrated into **PyTorch DDP**, achieving line rate. [Artifact open sourced at Github](#) [🔗](#).
- Evaluated across four LLM supervised fine-tuning workloads, achieving 34% end-to-end acceleration over MXFP8.

Research Assistant, Prediction-based MoE Offloading for LLM Serving

University College London

PRED-MoE. To appear at SIGCOMM '26 as a poster.

Feb 2026 - Today

- Designed PRED-MoE, a prediction-enhanced eviction policy for MoE expert offloading under limited GPU memory.
- Built a preliminary **vLLM**-based offloader plugin in python, with lightweight expert prediction machine learning model, showing up to $3.02\times$ higher end-to-end serving throughput over vLLM's state-of-the-art offloader.

Software Engineer Intern, Data Center Networking

Beijing, China

ByteDance, Publication H@NSDI.

Sept 2020 – Mar 2021

- Worked on developing Tiara, a P4+FPGA+CPU based stateful L4 load balancing system, achieving 1.6Tbps throughput while supporting 80M concurrent flows.
- Designed and implemented a high-performance (lock-free) control plane using DPDK+C making LB decisions and offloading connection tables to FPGA-NICs, achieving 4Mops per CPU core and microsecond-level latency.
- Collaborated with cross-functional hardware and software teams to deploy Tiara in production data centers.

OTHER RESEARCH EXPERIENCES

Research Assistant, Distributed Programmable Networks

PKU and UCL

Fast Reaction Algorithms for Network Coordination In Switches. Manuscript E.

June 2021 – Feb 2023

- o Built **FRANCIS**, a P4-based framework that facilitates running message-passing algorithms (MPA) in distributed programmable switches to achieve fast reaction to network events. [Artifact open-sourced at Github](#). [🔗](#)
- o Developed **FRANCIS** prototypes for real-world use cases, namely clock synchronization, multicast, and routing.
- o Delivered $100 \times \mu\text{s}$ reaction time ($18\times$ faster than state-of-the-arts), improving application performance.

Research Assistant, Network Measurement Algorithms

Peking University

Double Anonymous Sketch. Publication F@SIGMOD.

Mar 2020 – Aug 2020

- o Developed a generic "strong-unbiased" *Double Anonymous Sketch* achieving fairness in global Top-K flow detection.
- o Conducted extensive simulations and achieved significantly higher F1 Score than Waving Sketch and USS.

Research Assistant. Programmable in-network Caching.

University College London

The SQUID project. Publication G@CoNEXT.

Sept 2022 – Mar 2023

- o Proposed a data plane (P4) algorithm atop **SQUID**, a sampled quantiles q-MAX algorithm, for in-network caching systems—being the first to support a wide spectrum of caching policies and achieve real-time cache update.
- o Implemented a prototype of **SQUID-P4** and demonstrated that it achieves a near-optimal cache-hit ratio.

SKILLS

- o **Programming Languages:** C/C++, Python, Golang, CUDA/GPU programming.
- o **Software Engineering:** Distributed systems, Kubernetes, Docker, Bash, CMake/Makefile, Git.
- o **GPU and ML Systems:** CUDA optimization, Nsight profiling, NCCL, PyTorch, vLLM, Huggingface, lm-eval.
- o **LLM Infrastructure:** Distributed LLM training, LLM serving and deployment, gradient quantization, KV-cache transfer, Mixture-of-Expert offloading, prefix caching, speculative decoding, disaggregated prefill/decode.
- o **Networked systems:** DPDK, RDMA, P4, FPGA-NIC offload, data-center load balancing.

PUBLICATIONS AND MANUSCRIPTS

A. Beyond Throughput and Compression Ratios: Towards High E2E Utility of Gradient Compression [🔗](#)

Wenchen Han, Shay Vargaftik, Michael Mitzenmacher, Brad Karp, Ran Ben Basat. In HotNets '24.

B. DynamiQ: Accelerating Gradient Synchronization using Compressed Multi-hop All-reduce [🔗](#)

Wenchen Han, Shay Vargaftik, Michael Mitzenmacher, Ran Ben Basat. In SIGCOMM '26 (Regular paper).

C. Lynx: Progressive Speculative Quantization for accelerating KV Transfer in Long-Context Inference [🔗](#)

Wenchen Han, Gingfung Matthew Yeung, Marco Barletta, William Toner, Amory Hoste, Adam Barker. Under re-submission.

D. PRED-MoE: Prediction-Enhanced Expert Prefetching and Eviction for MoE Offloading [🔗](#)

Wenchen Han, Shay Vargaftik, Michael Mitzenmacher, Ran Ben Basat. In SIGCOMM '26 (Poster).

E. FRANCIS: Fast Reaction Algorithms for Network Coordination In Switches [🔗](#)

Wenchen Han, Vic Feng, Gregory Schwartzman, Yuliang Li, Michael Mitzenmacher, Minlan Yu, Ran Ben Basat.

F. Double-Anonymous Sketch: Achieving Fairness for Finding Global Top-K Frequent Items [🔗](#)

(Co-first author) Yikai Zhao*, **Wenchen Han***, Zheng Zhong*, Yinda Zhang, Tong Yang, Bin Cui. In SIGMOD 2023.

G. SQUID: Faster Analytics via Sampled Quantiles Data-structure [🔗](#)

(Alphabetical order) Ran Ben Basat, Gil Einziger, **Wenchen Han**, Bilal Tayh. In CoNEXT 2024.

H. Tiara: A Scalable and Efficient Hardware Acceleration Architecture for Stateful Layer-4 Load Balancing [🔗](#)

Chaoliang Zeng, Layong Luo, Teng Zhang, Zilong Wang, Luyang Li, **Wenchen Han**, Nan Chen, Lebing Wan, Lichao Liu, Zhipeng Ding, Xiongfei Geng, Tao Feng, Feng Ning, Kai Chen, Chuanxiong Guo. In NSDI '22

I. Achieving Top-K-fairness for Finding Global Top-K Frequent Items [🔗](#)

Yikai Zhao, Wei Zhou, **Wenchen Han**, Zheng Zhong, Yinda Zhang, Xiuqi Zheng, Tong Yang, Bin Cui. In TKDE 2025

J. Bounded Memory in Distributed Networks [↗](#)

(Alphabetical order) Ran Ben Basat, Keren Censor-Hillel, Yi-Jun Chang, **Wenchen Han**, Dean Leitersdorf, Gregory, Schwartzman. In SPAA 2025.

TALKS

- **Towards High End-to-end Utility of Gradient Compression for AI Training.** In *HotNets 2024*.
- **Reduced Communication Is not All You Need: Towards High End-to-end Utility of Gradient Compression.** In *Coseners 2024*.
- **FRANCIS: Fast Reaction Algorithms for Network Coordination In Switches.** In *Coseners 2023*.

HONORS AND AWARDS

- **John Hopcroft Turing Class Award**, PKU (8th / 58) *Oct. 2021*
- **John Hopcroft Turing Class Award**, PKU (11th / 58) *Nov. 2020*
- **Silver Medals** in APIO 2017 and NOI 2017.

PROFESSIONAL SERVICES

- **CoNEXT '25** shadow TPC.